

① a) f_1 is linear.

- Scaling by 2 duplicates the output

- When $x = \emptyset$, $y = \emptyset$

f_2 is not linear

$$\rightarrow f(2x) \neq 2f(x)$$

$$\rightarrow f(-x) \neq -f(x)$$

$$\rightarrow f(\emptyset) \neq \emptyset$$

f_3 is linear

$$\rightarrow f(2x) = 2f(x)$$

$$\rightarrow f(-x) = -f(x)$$

$$\rightarrow f\left(\frac{x}{2}\right) = \frac{1}{2}f(x)$$

$$\rightarrow f(\emptyset) = \emptyset$$

f_4 is linear

$$\rightarrow f(2x) = 2f(x)$$

$$\rightarrow f(-x) = -f(x)$$

$$\rightarrow f\left(\frac{x}{2}\right) = \frac{1}{2}f(x)$$

$$\rightarrow f(\emptyset) = \emptyset$$

f_5 is not linear

$$\rightarrow f(2x) \neq 2f(x)$$

$$\rightarrow f\left(\frac{x}{2}\right) \neq \frac{1}{2}f(x)$$

f_6 is not linear

$$\rightarrow f(-x) \neq -f(x)$$

b) f_1 : $f_1(1) = 1.8$, $f_1(\emptyset) = \emptyset$
 $f_2(1) = 0.5$, $f_2(\emptyset) = \emptyset$

$$m_1 = \frac{1.8}{1} = 1.8 \rightarrow f_1(x) = 1.8x$$

$$m_2 = \frac{0.5}{1} = 0.5 \rightarrow f_2(x) = 0.5x$$

$$f_3 : f_1(\emptyset) = \emptyset, f_1(1) = 0.5 \therefore f_1(x) = 0.5x$$

$$f_2(\emptyset) = \emptyset, f_2(1) = 0.5 \therefore f_2(x) = 0.5x$$

$$f_1: \begin{aligned} f_1(\emptyset) &= \emptyset, & f_1(1) &= 0.2 & \therefore & f_1(x) = 0.2x \\ f_2(\emptyset) &= \emptyset, & f_2(1) &= 1.4 & \therefore & f_2(x) = 1.4x \end{aligned}$$

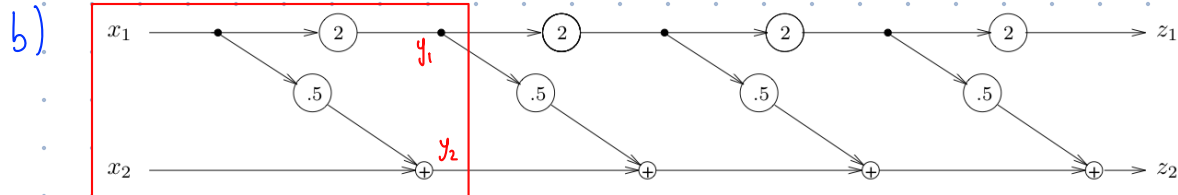
c) It is f_6

$$f\left(\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}\right) \neq f\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) + f\left(\begin{bmatrix} -1 \\ -1 \end{bmatrix}\right)$$

$$f\left(\begin{bmatrix} \emptyset \\ \emptyset \end{bmatrix}\right) = \emptyset \neq \begin{bmatrix} 1.5 \\ 1.3 \end{bmatrix} + \begin{bmatrix} -2.3 \\ -0.6 \end{bmatrix} = \begin{bmatrix} -0.8 \\ 0.7 \end{bmatrix}$$

$$f_6(x_1, x_2) = \begin{cases} 1.5x_1, 1.3x_2 & \text{if } x_1 > \emptyset, x_2 > \emptyset \\ -1.3x_1, 1.4x_2 & \text{if } x_1 < \emptyset, x_2 > \emptyset \\ -2.3x_1, -0.6x_2 & \text{if } x_1 < \emptyset, x_2 < \emptyset \\ 0.5x_1, 0.86x_2 & \text{if } x_1 > \emptyset, x_2 < \emptyset \\ \emptyset, \emptyset & \text{otherwise} \end{cases}$$

② a) $y_1 = 2x_1$
 $y_2 = 0.5x_1 + x_2$ $A = \begin{bmatrix} 2 & \emptyset \\ 0.5 & 1 \end{bmatrix}$



i) $z_1 = 8y_1$
 $z_2 = y_2 + 0.5y_1 + y_1 + 2y_1$ $B = A^4 = \begin{bmatrix} 2 & \emptyset \\ 0.5 & 1 \end{bmatrix}^4 = \begin{bmatrix} 16 & \emptyset \\ 7.5 & 1 \end{bmatrix}$

ii) $z_1 = 16x_1$
 $z_2 = 0.5x_1 + x_2 + x_1 + 2x_1 + 4x_1 = 7.5x_1 + x_2$, $B = \begin{bmatrix} 16 & \emptyset \\ 7.5 & 1 \end{bmatrix}$

③ Defining $A = [f(e_1) \ f(e_2) \ \dots \ f(e_n)] \in \mathbb{R}^{m \times n}$

where $e_1, \dots, e_n$ are the standard basis vectors of $\mathbb{R}^n$

and A_{ij} is the i -th component of $f(e_j)$

Then $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$, so $x = x_1 e_1 + \dots + x_n e_n$

By linearity: $f(x) = f(x_1 e_1 + \dots + x_n e_n) = x_1 f(e_1) + \dots + x_n f(e_n)$

Which is the same as $f(x) = Ax$

④ a) $f(\alpha x + \beta y) = \underline{A(\alpha x + \beta y) + b}$

$$\alpha f(x) + \beta f(y) = \alpha(Ax + b) + \beta(Ay + b) = A(\alpha x + \beta y) + \underbrace{(\alpha + \beta)}_1 b$$

$$= \underline{A(\alpha x + \beta y) + b}$$

So $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$

b) $\rightarrow$ First show $g(\alpha x) = \alpha g(x)$

From a), using $y = \emptyset$, $\beta = 1 - \alpha$:

$$f(\alpha x + (1 - \alpha)\emptyset) = f(\alpha x) = \alpha f(x) + (1 - \alpha)f(\emptyset)$$

$$g(\alpha x) = f(\alpha x) - f(\emptyset) = \alpha f(x) + (1 - \alpha)f(\emptyset) - f(\emptyset) = \alpha f(x) - \alpha f(\emptyset)$$

$$= \alpha(f(x) - f(\emptyset))$$

So $g(\alpha x) = \alpha g(x)$

$\rightarrow$ Now show $g(x + y) = g(x) + g(y)$

$$g(x + y) = 2g\left(\frac{x+y}{2}\right) = 2\left[f\left(\frac{x+y}{2}\right) - f(\emptyset)\right] = 2\left[\frac{1}{2}f(x) + \frac{1}{2}f(y) - f(\emptyset)\right]$$

$$= f(x) + f(y) - 2f(\emptyset) = \underbrace{(f(x) - f(\emptyset))}_{g(x)} + \underbrace{(f(y) - f(\emptyset))}_{g(y)}$$

So $g(x + y) = g(x) + g(y)$

Now, since $g(\cdot)$ is linear, there exists an A such that $g(x) = Ax$

so $g(x) = f(x) - f(\emptyset) = Ax$

$$f(x) = Ax + f(\emptyset)$$

Let $b = f(\emptyset)$, then

$$f(x) = Ax + b$$

⑤ a) $Z = BY$, where $B_{ij} = 0$ for $j < i$ (upper triangular)

b) $W = [v_2, v_1, v_4, v_3, \dots, v_n, v_{n-1}]$, where v_i is a vector

or

$$W = VP, \text{ where } P = \begin{bmatrix} 0 & 1 & & & \\ 1 & 0 & & & \\ & & 0 & 1 & \\ & & 1 & 0 & \\ & & & & \ddots \end{bmatrix}$$

c) $P^T Q > 0$

d) $\text{diag}(P^T Q) > 0$

e) $A = [A_1, A_2]$, where A_1 is the first k -cols
and A_2 the remaining

Then $A_1^T A_2 = 0$

⑥

$B = A^k$, B_{ij} is the number of paths of length k from node i to j ,

A path of length k means a sequence of nodes $i = v_0, v_1, \dots, v_k = j$

such that every consecutive pair is connected by an edge.

→ For $k=1$, A_{ij} counts paths of length 1

→ For $k=2$, $(A^2)_{ij} = \sum_{p=1}^n A_{ip} A_{pj}$, counting intermediate nodes p giving paths
 $i \rightarrow p \rightarrow j$

And the same extends for A^k

EE263 Homework 1

Summer 2026

- 1. Linear or not.** We have built an interactive tool with six mystery maps $f_1, \dots, f_6$, each from $\mathbb{R}^2$ to $\mathbb{R}^2$:

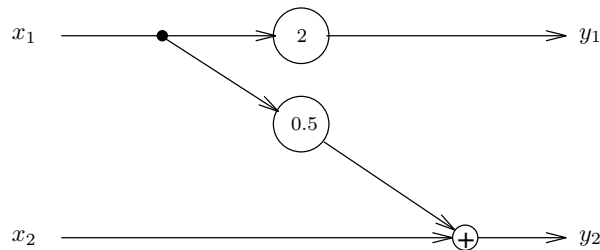
<https://amirafsharrad.github.io/teaching/ee263-matrix-methods/sim/linear-or-not/>

You cannot see how the maps are defined, but you can probe them. Pick a map, set an input x (type its two components, or drag the input vector with the mouse), and read off the output $y = f_i(x)$ in the second panel. You can also pin probes, so you can view f_i at several different inputs at the same time.

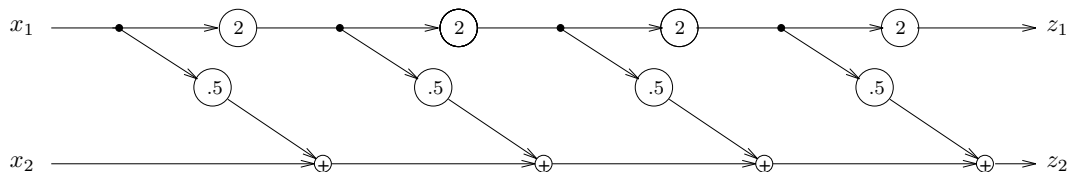
- For each of the six maps, decide whether it is linear. Explain how you reached each conclusion, citing the specific inputs and outputs you used.
- For each map you found to be linear, determine it exactly: give an explicit expression for f_i , precise enough that you (or we) could compute $f_i(x)$ for any input x without the tool. Explain the procedure you used.
- Exactly one of the non-linear maps is still homogeneous: it satisfies $f(\alpha x) = \alpha f(x)$ for every scalar α , yet it is not additive. Identify this map, justify your choice using the tool, and see whether you can determine (or guess) its definition.
- (Optional) Can you find, or at least guess, the definitions of the remaining non-linear maps?

2. Matrices and signal flow graphs.

- a) Find $A \in \mathbb{R}^{2 \times 2}$ such that $y = Ax$ in the system below:



- b) Find $B \in \mathbb{R}^{2 \times 2}$ such that $z = Bx$ in the system below:



Do this two ways: first, by expressing the matrix B in terms of A from the previous part (explaining why they are related as you claim); and second, by directly evaluating all possible paths from each x_j to each z_i .

3. (Optional) Representing linear functions as matrix multiplication. Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear. Show that there is a matrix $A \in \mathbb{R}^{m \times n}$ such that for all $x \in \mathbb{R}^n$, $f(x) = Ax$. (Explicitly describe how you get the coefficients A_{ij} from f , and then verify that $f(x) = Ax$ for any $x \in \mathbb{R}^n$.) Is the matrix A that represents f unique? In other words, if $\tilde{A} \in \mathbb{R}^{m \times n}$ is another matrix such that $f(x) = \tilde{A}x$ for all $x \in \mathbb{R}^n$, then do we have $\tilde{A} = A$? Either show that this is so, or give an explicit counterexample.

4. Affine functions. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is called *affine* if for any $x, y \in \mathbb{R}^n$ and any $\alpha, \beta \in \mathbb{R}$ with $\alpha + \beta = 1$, we have

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y).$$

(Without the restriction $\alpha + \beta = 1$, this would be the definition of linearity.)

- a) Suppose that $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Show that the function $f(x) = Ax + b$ is affine.
- b) Now the converse: Show that any affine function f can be represented as $f(x) = Ax + b$, for some $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. (This representation is unique: for a given affine function f there is only one A and one b for which $f(x) = Ax + b$ for all x .)

Hint. Show that the function $g(x) = f(x) - f(0)$ is linear.

You can think of an affine function as a linear function, plus an offset. In some contexts, affine functions are (mistakenly, or informally) called linear, even though in general they are not. (Example: $y = mx + b$ is described as ‘linear’ in US high schools.)

5. Express the following statements in matrix language. You can assume that all matrices mentioned have appropriate dimensions. Here is an example: “Every column of C is a linear combination of the columns of B ” can be expressed as “ $C = BF$ for some matrix F ”.

There can be several answers; one is good enough for us.

- a) Suppose Z has n columns. For each i , row i of Z is a linear combination of rows $i, \dots, n$ of Y .
- b) W is obtained from V by permuting adjacent odd and even columns (*i.e.*, 1 and 2, 3 and 4, $\dots$).
- c) Each column of P makes an acute angle with each column of Q .
- d) Each column of P makes an acute angle with the corresponding column of Q .
- e) The first k columns of A are orthogonal to the remaining columns of A .

6. Counting paths in an undirected graph. Consider an undirected graph with n nodes, and no self loops (*i.e.*, all branches connect two different nodes). Let $A \in \mathbb{R}^{n \times n}$ be the *node adjacency matrix*, defined as

$$A_{ij} = \begin{cases} 1 & \text{if there is a branch from node } i \text{ to node } j \\ 0 & \text{if there is no branch from node } i \text{ to node } j \end{cases}$$

Note that $A = A^\top$, and $A_{ii} = 0$ since there are no self loops. We can interpret A_{ij} (which is either zero or one) as the number of branches that connect node i to node j . Let $B = A^k$, where $k \in \mathbb{Z}$, $k \geq 1$. Give a simple interpretation of B_{ij} in terms of the original graph. (You might need to use the concept of a *path* of length m from node p to node q .)

7. (Optional) Some matrices from signal processing. We consider $x \in \mathbb{R}^n$ as a signal, with x_i the (scalar) value of the signal at (discrete) time period i , for $i = 1, \dots, n$. Below we describe several transformations of the signal x , that produce a new signal y (whose dimension varies). For each one, find a matrix A for which $y = Ax$.

- a) $2\times$ *up-conversion with linear interpolation*. We take $y \in \mathbb{R}^{2n-1}$. For i odd, $y_i = x_{(i+1)/2}$. For i even, $y_i = (x_{i/2} + x_{i/2+1})/2$. Roughly speaking, this operation doubles the sample rate, inserting new samples in between the original ones using linear interpolation.
- b) $2\times$ *down-sampling*. We assume here that n is even, and take $y \in \mathbb{R}^{n/2}$, with $y_i = x_{2i}$.
- c) $2\times$ *down-sampling with averaging*. We assume here that n is even, and take $y \in \mathbb{R}^{n/2}$, with $y_i = (x_{2i-1} + x_{2i})/2$.

8. (Optional) Gradient of some common functions. Recall that the gradient of a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, at a point $x \in \mathbb{R}^n$, is defined as the vector

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix},$$

where the partial derivatives are evaluated at the point x . The first order Taylor approximation of f , near x , is given by

$$\hat{f}_{\text{tay}}(z) = f(x) + \nabla f(x)^\top (z - x).$$

This function is affine, *i.e.*, a linear function plus a constant. For z near x , the Taylor approximation $\hat{f}_{\text{tay}}$ is very near f . Find the gradient of the following functions. Express the gradients using matrix notation.

- a) $f(x) = a^\top x + b$, where $a \in \mathbb{R}^n$, $b \in \mathbb{R}$.
- b) $f(x) = x^\top Ax$, for $A \in \mathbb{R}^{n \times n}$.
- c) $f(x) = x^\top Ax$, where $A = A^\top \in \mathbb{R}^{n \times n}$. (Yes, this is a special case of the previous one.)